

SHIELDED CHAMBER DESIGN SPECIFICATIONS

For Consciousness-Dependent Quantum Decoherence Measurements

Phase 0 Configuration

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The Silence Paradigm Project

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1. Design Objective

This document specifies the engineering requirements for a shielded experimental chamber designed to test whether the cognitive state of a conscious observer produces measurable changes in the decoherence rate (T_2) of a nearby quantum system. The chamber creates a controlled environment of maximum electromagnetic, acoustic, vibrational, and thermal isolation while maintaining a symmetric geometry optimized for minimal boundary-induced decoherence.

The independent variable is the observer's cognitive state (seven controlled conditions, verified by real-time EEG). The dependent variable is the coherence time T_2 of the quantum sensor. The chamber must reduce all known environmental coupling channels below the threshold where standard decoherence models predict T_2 limits, such that any residual variation across consciousness conditions cannot be attributed to uncontrolled environmental factors.

1.1 Performance Requirements

- Electromagnetic attenuation: >120 dB across 10 Hz – 10 GHz
- Magnetic field attenuation: >60 dB for DC and low-frequency fields
- Acoustic isolation: >80 dB; interior ambient <20 dB SPL
- Vibration isolation: <10⁻⁶ m/s² across 1–100 Hz at sensor platform
- Thermal stability: ±0.01 K over 60-minute session
- Interior geometry: Perfect cube, 3.0 m × 3.0 m × 3.0 m clear volume
- Observer capacity: Single seated adult at geometric center

2. Overall Architecture

2.1 Dimensions

Parameter	Specification
Inner clear volume	3.0 m × 3.0 m × 3.0 m (27.0 m³)
Outer footprint	4.2 m × 5.4 m × 4.2 m (incl. airlock extension on east face)
Total shielding thickness	0.6 m per side (all axes)
Orientation	Entry on east face; observer faces west when seated
Primary access	East face — dual-door airlock entry system (see Section 3)
Outer door	1.0 m W × 2.1 m H, east face of airlock
Inner door	1.0 m W × 2.1 m H, west face of airlock (enters main chamber)
Door seals (both)	Multi-layer: RF gasket + mu-metal overlap + acoustic compression seal
Feedthrough panel	West face (opposite entry), centered
Total chamber mass (est.)	9,000 – 13,000 kg (incl. airlock)
Foundation requirement	Reinforced concrete slab, 300 mm min, vibration-isolated

2.2 Geometry Rationale

The perfect cubic geometry is selected for four engineering reasons:

- **Spatial symmetry:** Minimizes anisotropic boundary conditions that could produce position-dependent decoherence variation
- **Mode density:** Cubic geometry supports the lowest and most uniform electromagnetic mode density for a given volume
- **Fabrication simplicity:** Flat panels simplify shielding layer construction, seam welding, and quality assurance
- **East-facing entry:** The entry on the east face positions the observer facing west (toward the feedthrough panel / deepest interior), maximizing the distance between the access opening (highest seal-leakage risk) and the primary sensor array. The observer walks from east to west through the airlock into the main chamber, passing through progressive isolation layers sequentially

2.3 Coordinate System

Origin at inner floor corner (southeast). X-axis west, Y-axis north, Z-axis vertical. Geometric center at (1.5, 1.5, 1.5) m. The observer enters from the east (X = 0) and is seated at center facing west (X = 3.0). The feedthrough panel is on the west face (X = 3.0). All sensor and observer positions referenced to this coordinate system.

3. Airlock Entry System

A dual-door airlock anteroom is positioned on the east face of the main chamber. The airlock serves three critical engineering functions: (1) it prevents direct line-of-sight RF leakage when either door is open, (2) it provides a contamination buffer where the observer’s electromagnetic emissions from clothing, devices, and body can be checked and stabilized before entering the shielded interior, and (3) it ensures the observer passes sequentially through all five isolation layers, with no bypass path.

3.1 Airlock Dimensions

Parameter	Specification
Interior volume	1.0 m (E–W) × 1.2 m (N–S) × 2.3 m (height)
Outer door (east face)	1.0 m W × 2.1 m H, opens outward
Inner door (west face)	1.0 m W × 2.1 m H, opens inward to main chamber
Interlock	Only one door can be open at a time (electrical interlock + mechanical latch)
Shielding	All five layers continuous through airlock walls, ceiling, and floor
Lighting	Battery-powered LED (shielded), manual switch inside airlock
Communication	Fiber-optic intercom to control room, active in airlock

3.2 Entry Protocol

The following standardized entry sequence is followed for every session:

- **Step 1 — Pre-entry check:** Observer removes all metallic objects, electronic devices, and magnetically contaminated clothing in the preparation area outside the chamber. EEG cap is fitted and verified.
- **Step 2 — Outer door open:** Observer enters airlock. Outer door is sealed behind the observer.

- Step 3 — Stabilization: Observer remains in airlock for 60 seconds. RF power meter inside airlock confirms observer's residual electromagnetic emission is below threshold (<-80 dBm across 10 kHz–10 GHz). Fluxgate magnetometer confirms no ferromagnetic contamination.
- Step 4 — Inner door open: Once contamination check passes, inner door is released. Observer enters main chamber and proceeds to seated position at geometric center.
- Step 5 — Inner door sealed: Inner door is closed and sealed. Both doors now closed. Full shielding integrity restored.
- Step 6 — Settling period: 10-minute settling period begins (data discarded). EEG baseline established.

3.3 Exit Protocol

Reverse sequence: inner door opens first (observer exits to airlock), inner door sealed, outer door opens, observer exits. At no point are both doors open simultaneously. Emergency override (panic bar) on inner door bypasses interlock for immediate egress.

Design Note: Sequential Layer Passage

The airlock ensures that the observer's entry path passes through every isolation layer in sequence: Layer 5 (outer Faraday) → Layer 4 (vibration) → Layer 3 (RF copper) → Layer 2 (mu-metal) → Layer 1 (acoustic/thermal) → interior. No layer is bypassed. This is critical for maintaining shielding integrity and preventing contamination bridging between the exterior environment and the sensor volume.

4. Five-Layer Isolation System

The isolation system consists of five concentric layers, each addressing a specific noise channel. Layers are numbered from innermost (Layer 1) to outermost (Layer 5). Total thickness per side: 600 mm.

Layer	Function	Material	Thickness	Attenuation
1 (Inner)	Acoustic + Thermal	Closed-cell acoustic foam + aerogel blanket	150 mm	>40 dB acoustic, ±0.01 K
2	Magnetic shielding	Mu-metal (80% Ni alloy)	3 mm	>60 dB DC/LF magnetic
3	RF/Microwave shielding	OFHC copper sheet, welded seams	2 mm	>100 dB (10 kHz–10 GHz)
4	Vibration isolation	Pneumatic isolators + neoprene dampers	200 mm	<10 m/s ² (1–100 Hz)
5 (Outer)	Structural + Faraday	Aluminum shell + copper mesh cage	245 mm	>120 dB combined EM

Layer ordering rationale: Acoustic/thermal innermost to establish the stable environment the observer and sensors experience. Magnetic shielding next to protect the quantum sensor from low-frequency field drift. RF shielding third to eliminate broadband electromagnetic interference. Vibration isolation fourth to mechanically decouple the inner assembly from the building. Structural/Faraday outermost to provide mechanical rigidity and a ground plane.

5. Layer-by-Layer Material Specifications

5.1 Layer 1 — Acoustic and Thermal Isolation

Component	Material	Specification	Supplier Type
Acoustic panels	Melamine closed-cell foam (BASF Basotect or equiv.)	100 mm thick, NRC >0.95, Class A fire rating	Acoustic specialty
Thermal blanket	Silica aerogel blanket (Aspen Aerogels Cryogel or equiv.)	50 mm thick, thermal conductivity <0.015 W/mK	Thermal insulation
Inner liner	Perforated aluminum sheet	1 mm thick, 20% open area, over acoustic foam	Sheet metal
Mounting	Non-magnetic stainless clips	316L SS, bonded to foam with silicone adhesive	Hardware

Installation: Aerogel blanket applied directly to inner face of Layer 2. Acoustic foam panels mounted over aerogel with air gap of 10 mm maintained by standoff clips. Perforated liner installed as inner surface for mechanical protection and uniform acoustic absorption.

5.2 Layer 2 — Magnetic Shielding

Component	Material	Specification	Notes
Primary shield	Mu-metal (MuShield or Magnetic Shield Corp.)	3 mm sheet, $\mu_r > 80,000$ initial, > 300,000 max	Must be hydrogen-annealed after forming

Component	Material	Specification	Notes
Seam treatment	Mu-metal overlap strips	50 mm overlap at all joints, spot-welded	Continuous magnetic path required
Penetrations	Mu-metal waveguide-below-cutoff tubes	ID 25 mm, length 250 mm minimum	All cable/fiber feedthroughs
Door overlap	Mu-metal finger stock + overlap flange	100 mm overlap on all four edges	Spring-loaded contact

CRITICAL: Hydrogen Annealing

All mu-metal components **MUST** be hydrogen-annealed at 1,150°C after final forming and before installation. Mechanical stress from cutting, bending, or drilling degrades permeability by up to 90%. Annealing restores full magnetic shielding performance. Budget 4–6 weeks lead time for annealing service.

5.3 Layer 3 — RF / Microwave Shielding

Component	Material	Specification	Notes
Shield panels	OFHC copper sheet (C10100)	2 mm thick, oxygen-free high-conductivity	Conductivity >100% IACS
Seam joining	Continuous TIG or silver-solder	All seams welded; no mechanical-only joints	RF leakage at gaps
Ground bonding	Copper braid to building ground	25 mm ² minimum cross-section	Single-point ground
Penetrations	Waveguide-below-cutoff copper tubes	ID 20 mm, length 200 mm	Bonded to copper shell
Door contact	BeCu RF finger stock gasket	Continuous around full door perimeter	Double row for redundancy

5.4 Layer 4 — Vibration Isolation

Component	Material	Specification	Notes
Active isolators	Pneumatic air-spring isolators (TMC or Newport)	4–8 units, 2,000 kg capacity each	Active feedback <1 Hz
Passive dampers	Neoprene/rubber pads	Shore A 40–60, 50 mm thick	Between frame and floor
Isolation frame	Welded steel frame	100 mm × 100 mm box section	Supports all inner layers
Sensor sub-platform	Granite or cast-iron optical table	1.5 m × 1.0 m × 0.3 m, mass >500 kg	Separate isolators from main
Flexible couplings	Bellows / flex conduits	All connections between inner and outer	Prevent vibration bridging

5.5 Layer 5 — Structural Shell and Outer Faraday Cage

Component	Material	Specification	Notes
Structural shell	6061-T6 aluminum plate	5 mm thick, welded frame	Provides mechanical envelope
Faraday mesh	Copper mesh, 200 × 200 per inch	Bonded to inner face of aluminum shell	Soldered at all overlaps

Component	Material	Specification	Notes
Structural frame	Aluminum extrusion	80 mm × 80 mm T-slot profile	Modular assembly
Floor	12 mm aluminum plate on steel frame	Level to ±1 mm over 4.2 m	Anti-static coating
Access panel	Removable panel opposite door	For equipment installation	Sealed after setup

6. Internal Configuration

6.1 Observer Station

Parameter	Specification
Position	Geometric center: (1.5, 1.5, 0.0) m, seated height places head at ~ (1.5, 1.5, 1.2) m
Seating	Non-magnetic meditation platform (wood/carbon fiber), no metal fasteners
EEG system	Wireless 64-channel cap (BrainProducts LiveAmp or equiv.), optical fiber data link
Physiological	Heart rate variability sensor, respiration band, skin conductance (all non-magnetic)
Comfort	Temperature-controlled seat pad, adjustable headrest, emergency call button
Session duration	45 min active condition + 25 min baseline = 70 min total occupancy

6.2 Quantum Sensor Array

Parameter	Specification
Primary sensor (Phase 0)	Nitrogen-vacancy (NV) center in diamond, room-temperature T_2 measurement
Primary sensor (Phase 1)	Superconducting transmon qubit, dilution refrigerator at 15 mK
Mounting	Non-magnetic optical table at center, 1.0 m above floor level
Distance to observer	1.0–1.5 m (adjustable along X-axis)
Secondary sensors	4× commercial QRNGs (ComScire PQ32MU), USB via optical fiber
Exploratory	Single-photon detectors (200–1000 nm), <100 dark counts/s

6.3 Environmental Monitoring Array

Instrument	Sensitivity	Sampling Rate	Purpose
3-axis fluxgate magnetometer	<10 nT resolution	1 kHz	Residual magnetic field monitoring
SQUID magnetometer	<10 fT/√Hz	10 kHz	Ultra-sensitive magnetic monitoring
RF power meter	−80 dBm sensitivity	Continuous	Residual RF leakage detection
3-axis accelerometer	10 ^{−9} g resolution	1 kHz	Vibration monitoring at sensor
Pt100 RTD sensors (×4)	±0.001 K resolution	10 Hz	Temperature at 4 quadrants
Humidity sensor	±0.5% RH	1 Hz	Interior humidity stability
Barometric sensor	±0.1 hPa	1 Hz	Atmospheric pressure drift

All environmental channels are recorded synchronously with quantum sensor data and EEG via a common time base (GPS-disciplined 10 MHz reference, <1 μs inter-channel synchronization). Any anomaly in quantum data triggers automatic cross-correlation with all environmental channels.

7. Support Systems

7.1 Power

- No AC mains inside chamber
- Battery bank (48V LiFePO₄, 5 kWh) in outer equipment room
- DC-DC converters with optical isolation for each subsystem
- All power lines filtered through EMI/RFI feedthrough filters at Layer 3 boundary
- UPS backup for data acquisition (minimum 2 hours runtime)

7.2 Air Handling

- Closed-loop HEPA-filtered recirculation, no external air exchange during sessions
- CO₂ scrubber (soda lime or molecular sieve), capacity for 2-hour sessions
- Flow rate <0.1 m/s at observer position to minimize turbulent acoustic noise
- Ducting through vibration-isolated bellows with acoustic baffles
- O₂ monitoring with automatic abort if O₂ < 19.5%

7.3 Data Acquisition

- All signals routed via single-mode fiber-optic feedthroughs (FC/APC connectors)
- Fiber penetrations through waveguide-below-cutoff tubes in Layers 2, 3, and 5
- External DAQ system: National Instruments PXIe chassis with multi-channel digitizers
- Common time base: GPS-disciplined 10 MHz oscillator, distributed to all subsystems
- Data storage: Redundant NVMe RAID array, minimum 10 TB

7.4 Safety

- Emergency egress: Door operable from inside without tools (panic bar)
- Redundant O₂ monitoring with audible/visual alarm inside and outside
- External video monitoring via fiber-optic camera (infrared, no EM emission inside chamber)
- Two-way fiber-optic intercom
- Automatic session abort on any safety threshold breach
- Fire suppression: Inert gas (Ar/N₂) system, manual trigger only

8. Quantum Sensor Integration

8.1 NV-Center in Diamond (Phase 0)

Parameter	Specification
Sensor type	Single NV center or NV ensemble in CVD diamond
T ₂ baseline	1–10 ms at room temperature (spin echo)
Optical excitation	532 nm laser, fiber-coupled, <10 mW at diamond
Detection	Confocal fluorescence collection, 637–800 nm, APD detector
Microwave control	Coplanar waveguide on diamond mount, ~2.87 GHz
Readout rate	~10 ³ T ₂ measurements per session
Mounting	Diamond on PCB mount, on optical table, thermally stabilized
Advantages	Room temperature, no cryogenics, moderate cost, well-characterized
Limitations	Lower sensitivity to environmental perturbation than superconducting qubit

8.2 Superconducting Transmon Qubit (Phase 1 Upgrade)

Parameter	Specification
Sensor type	Transmon qubit in 3D aluminum cavity
T ₂ baseline	50–200 μs at 15 mK
Operating temp	15 mK (dilution refrigerator required)
Readout	Dispersive readout via microwave cavity, HEMT amplifier chain
Cryostat	Dry dilution refrigerator (BlueFors LD or equiv.)
Footprint	1.0 m × 1.0 m floor, 2.0 m height (including vibration frame)
Cryogenic lines	Through dedicated vibration-isolated feedthroughs
Advantages	Maximum sensitivity to environmental perturbation
Limitations	Cost (~€200K–500K for cryostat), complexity, specialized operators

8.3 QRNG Array

Parameter	Specification
Units	4× ComScire PQ32MU (32 Mbps each)
Interface	USB via fiber-optic USB extender
Mounting	Non-magnetic rack adjacent to optical table
Data volume	~14.4 GB per unit per 60-min session
Analysis	Entropy, autocorrelation, mean deviation; NIST SP 800-22 suite

9. Mechanical Drawings (Descriptive)

Formal CAD drawings to be produced in SolidWorks or Fusion 360 during detailed design phase. The following text descriptions define the required drawing set.

9.1 Drawing List

Drawing No.	Title	View	Content
TSP-001	Chamber Assembly — Plan View	Top-down, Z = 1.5 m section	All 5 layers, airlock, feedthroughs, observer and sensor positions
TSP-002	Chamber Assembly — Elevation	South-facing section at Y = 1.5 m	Layer stack-up, floor structure, ceiling, height dimensions
TSP-003	Chamber Assembly — Cross Section	West-facing section at X = 1.5 m	Entry axis view, airlock detail, cable routing
TSP-004	Airlock Assembly	Plan + section detail	Dual-door arrangement, interlock, RF check station, seal details
TSP-005	Layer 2 — Mu-Metal Shell	Exploded isometric	Panel layout, seam overlaps, annealing callouts, penetrations
TSP-006	Layer 3 — Copper Shell	Exploded isometric	Panel layout, weld seam map, grounding, RF gasket details
TSP-007	Layer 4 — Isolation Platform	Plan + elevation	Isolator positions, frame structure, flexible coupling routes
TSP-008	Internal Layout	Isometric cutaway	Observer station, optical table, sensor mounts, cable routing
TSP-009	Door Assemblies (Inner + Outer)	Section detail	Multi-layer seal stack, hinge, latch, panic bar, interlock wiring
TSP-010	Feedthrough Panel (West Face)	Section detail	Waveguide-below-cutoff tubes, fiber connectors, filter mounts
TSP-011	Foundation Plan	Plan view	Concrete slab, isolation trench, grounding grid, utilities

9.2 Plan View Description (TSP-001)

Top-down cross-section at Z = 1.5 m (mid-height). Shows concentric square layers from outside in: Layer 5 aluminum shell (outermost, dark gray), Layer 4 isolation frame (hatched), Layer 3 copper shell (copper-colored), Layer 2 mu-metal shell (light gray), Layer 1 acoustic/thermal panels (white). Airlock extension on east face shown with dual doors (outer and inner) and interlock annotation. Interior shows observer platform at center (circle, Ø 0.8 m), optical table 1.0 m west of center (rectangle, 1.5 m × 1.0 m), QRNG rack north of optical table. Feedthrough panel on west face (opposite entry). Observer entry path shown as dashed arrow from east through airlock to center. All dimensions in millimeters.

9.3 Elevation Description (TSP-002)

South-facing cross-section at Y = 1.5 m (mid-width). Shows floor assembly (foundation slab, isolation mounts, floor plate), five concentric layers rising vertically, ceiling assembly. Airlock visible on east side with dual doors. Observer seated figure (outline) at center facing west. Optical table with NV-center sensor at 1.0 m height, positioned west of observer. Overhead cable/fiber routing tray at 2.7 m. Air handling duct entry at ceiling level with acoustic baffle detail. Cryostat position indicated (Phase 1 dashed outline) with vibration isolation frame. All dimensions in millimeters.

10. Bill of Materials and Cost Estimates

All costs are estimates in EUR for Phase 0 configuration (NV-center sensor, no cryostat). Costs assume single-unit procurement; volume discounts may apply for some items. Supplier quotes required for final budget.

10.1 Shielding Materials

Item	Quantity	Unit Cost (€)	Total (€)	Notes
Mu-metal sheet, 3 mm	60 m ²	250/m ²	15,000	Incl. hydrogen annealing
OFHC copper sheet, 2 mm	60 m ²	120/m ²	7,200	C10100 grade
Aluminum plate, 5 mm	110 m ²	45/m ²	4,950	6061-T6
Copper mesh, 200/inch	110 m ²	25/m ²	2,750	Faraday layer
Acoustic foam panels	55 m ²	80/m ²	4,400	Basotect or equiv.
Aerogel blanket, 50 mm	55 m ²	150/m ²	8,250	Cryogel Z or equiv.
RF finger stock gasket	20 m	60/m	1,200	BeCu, door + panels
Mu-metal finger stock	10 m	90/m	900	Door magnetic seal

Subtotal — Shielding Materials: €44,650

10.2 Structural and Mechanical

Item	Quantity	Unit Cost (€)	Total (€)	Notes
Aluminum T-slot extrusion	240 m	15/m	3,600	80×80 mm profile (incl. airlock)
Pneumatic isolators	6 units	2,500 ea	15,000	TMC or Newport
Neoprene pads	12 units	150 ea	1,800	Floor isolation
Steel isolation frame	1 set	4,000	4,000	Welded box section
Optical table (granite)	1 unit	8,000	8,000	1.5 m × 1.0 m × 0.3 m
Outer door assembly (east)	1 unit	5,000	5,000	Multi-layer, outward-opening
Inner door assembly (east)	1 unit	6,000	6,000	Multi-layer, inward-opening
Door interlock system	1 unit	2,500	2,500	Electrical + mechanical, dual-door
Airlock shielding extension	1 set	4,500	4,500	All 5 layers continuous through airlock
Airlock RF check station	1 unit	1,500	1,500	RF meter + fluxgate, wall-mounted
Feedthrough panels	2 units	2,000 ea	4,000	West face, waveguide tubes + connectors
Hardware / fasteners	Lot	—	2,000	Non-magnetic where required

Subtotal — Structural: €52,400

10.3 Sensors and Instrumentation

Item	Quantity	Unit Cost (€)	Total (€)	Notes
NV-center diamond sensor kit	1 unit	15,000	15,000	Incl. optics + MW source
ComScire PQ32MU QRNG	4 units	1,350 ea	5,400	32 Mbps each
64-ch wireless EEG	1 system	12,000	12,000	BrainProducts or equiv.
SQUID magnetometer	1 unit	18,000	18,000	Single-channel, LN ₂ cooled
Fluxgate magnetometer	1 unit	3,000	3,000	3-axis
Single-photon detectors	2 units	6,000 ea	12,000	APD, 200–1000 nm
Pt100 RTD sensors	4 units	200 ea	800	0.001 K resolution
Accelerometer (3-axis)	1 unit	2,500	2,500	Seismic-grade
RF power meter	1 unit	1,500	1,500	Broadband

Subtotal — Sensors: €70,200

10.4 Support Systems

Item	Quantity	Unit Cost (€)	Total (€)	Notes
DAQ system (NI PXIe)	1 chassis + 4 cards	25,000	25,000	Multi-channel digitizer
GPS 10 MHz reference	1 unit	2,000	2,000	Time base
Battery bank (48V, 5 kWh)	1 unit	3,000	3,000	LiFePO ₄
DC-DC converters + filters	Lot	—	2,000	Optically isolated
HEPA air handler + CO ₂ scrub	1 system	5,000	5,000	Closed-loop
Fiber-optic USB extenders	6 units	400 ea	2,400	For QRNGs + EEG
Fiber-optic cables + connectors	Lot	—	1,500	Single-mode, FC/APC
Safety systems (O ₂ , intercom, camera)	Lot	—	3,000	Redundant
Workstation + storage	1 system	4,000	4,000	Control room

Subtotal — Support: €47,900

10.5 Labor and Integration

Item	Duration	Rate (£/day)	Total (€)	Notes
Chamber fabrication	30 days	500	15,000	Skilled metalworker + assistant
Shielding installation	15 days	600	9,000	Specialist RF/magnetic
Sensor integration	10 days	800	8,000	Quantum sensor specialist
DAQ integration + calibration	10 days	600	6,000	Systems engineer
Commissioning + verification	10 days	600	6,000	Full-system testing

Subtotal — Labor: €44,000

10.6 Budget Summary

Category	Cost (€)
Shielding Materials	44,650
Structural and Mechanical (incl. airlock)	52,400
Sensors and Instrumentation	70,200
Support Systems	47,900
Labor and Integration	44,000
Contingency (15%)	38,872
TOTAL PHASE 0 ESTIMATE	€298,022

Budget Note
This estimate of €298K falls within the Phase 0 range specified in the experimental protocol (€70K–€150K for minimal configuration, scaling to €300K+ for full instrumentation). The estimate above represents a fully instrumented Phase 0 with all environmental monitoring channels and the dual-door airlock entry system. A minimal Phase 0 (€70–100K) could be achieved by using a smaller 1.5 m cube with single door, omitting the SQUID magnetometer and single-photon detectors, and reducing the isolation platform to passive-only.

11. Fabrication Sequence

Total estimated fabrication timeline: 14–18 weeks from material procurement to first calibration session.

11.1 Phase Schedule

Phase	Duration	Activities	Dependencies
1. Procurement	4–6 weeks	Order mu-metal (+ annealing), copper, aluminum, isolators, sensors	Budget approval
2. Foundation	1–2 weeks	Pour/verify reinforced concrete slab, install grounding grid	Site access
3. Outer shell	2 weeks	Assemble Layer 5 aluminum frame and panels, install Faraday mesh	Foundation complete
4. Vibration system	1 week	Install isolation frame, pneumatic isolators, passive dampers	Outer shell complete
5. Copper shell	1–2 weeks	Install Layer 3 copper panels, weld seams, install feedthroughs	Isolation frame stable
6. Mu-metal shell	1–2 weeks	Install Layer 2 panels (annealed), overlap seams, penetration tubes	Copper shell complete
7. Acoustic/thermal	1 week	Install aerogel blanket, acoustic foam, perforated liner	Mu-metal complete
8. Door + airlock	1–2 weeks	Install airlock structure, dual doors, RF + magnetic + acoustic seals, interlock wiring	All layers complete
9. Internal fit-out	1–2 weeks	Optical table, sensor mounts, observer platform, cable routing	Chamber sealed
10. Systems integration	1–2 weeks	DAQ, power, air handling, safety, fiber routing	Internal fit-out
11. Commissioning	1–2 weeks	EM survey, magnetic field mapping, vibration spectra, T_2 baseline	Full integration

12. Integration and Commissioning

12.1 Shielding Verification

- Electromagnetic: Sweep 10 Hz – 10 GHz with calibrated signal generator outside, spectrum analyzer inside. Confirm >120 dB attenuation at all frequencies.
- Magnetic: DC field mapping with fluxgate at 27 grid points (3×3×3). Confirm residual field <50 nT. AC magnetic sweep 0.1–100 Hz, confirm >60 dB attenuation.
- Acoustic: Pink noise source outside at 90 dB SPL. Measure interior with calibrated microphone. Confirm <20 dB SPL interior.
- Vibration: Measure floor vibration spectra with and without active isolation engaged. Confirm <10⁻⁶ m/s² at sensor platform across 1–100 Hz.
- Thermal: 24-hour thermal recording with no occupant. Confirm ±0.01 K stability at sensor position.

12.2 Quantum Sensor Baseline

- NV-center: Measure T_2 (spin echo) with empty chamber. Establish baseline over minimum 100 measurements. Record mean, standard deviation, and drift.
- QRNG: Run all 4 units for 24 hours with empty chamber. Run NIST SP 800-22 test suite. Confirm all units pass all tests. Establish baseline entropy and autocorrelation statistics.
- Single-photon detectors: 24-hour dark count measurement. Confirm <100 counts/s. Characterize count rate stability.

12.3 System Integration Test

- Simultaneous recording of all channels (quantum sensor, QRNGs, EEG, all environmental monitors) for 60 minutes with empty chamber
- Verify time synchronization: <1 μ s inter-channel offset
- Verify data integrity: no dropped samples, no buffer overflows
- Verify artifact detection pipeline: inject known environmental perturbation and confirm cross-correlation detection

13. Phase Scaling Notes

Parameter	Micro-Pilot	Phase 0	Phase 1
Inner volume	1.5 m cube (portable)	3.0 m cube (full spec)	4.0–5.0 m cube
Location	Any quiet room	University lab	Underground ($\geq 1,000$ m)
Shielding	Portable mu-metal + copper box	Full 5-layer system	5-layer + rock overburden
Quantum sensor	None (QRNG only)	NV-center in diamond	Superconducting transmon
QRNG	1× ComScire PQ4000KU	4× ComScire PQ32MU	4× ComScire PQ32MU
EEG	Consumer-grade (optional)	64-channel wireless	256-channel wireless
Environmental monitors	Temperature only	Full array (7 channels)	Full array + SQUID
Budget	€2K–5K	€70K–290K	€500K–2M
Timeline	2–4 weeks	4–6 months	2–3 years
Goal	Demonstrate execution, preliminary data	Detect medium-large effects ($d > 0.3$)	Definitive test with full power

Appendix A: Performance Budget

The following table summarizes the expected contribution of each noise channel to T_2 limitation, and the isolation provided by the chamber design. The design target is that chamber-limited T_2 exceeds the intrinsic T_2 of the quantum sensor by at least 10 $\times$, ensuring that any observed T_2 variation is sensor-intrinsic or consciousness-state-dependent, not chamber-limited.

Noise Channel	Unshielded Level	Chamber Attenuation	Residual Level	T ₂ Impact
RF/Microwave (10 kHz–10 GHz)	~–30 dBm ambient	>120 dB	<–150 dBm	Negligible
DC magnetic field	~50 μ T (Earth field)	>60 dB	<50 nT	Below NV T ₂ limit
AC magnetic (50/60 Hz)	~100 nT typical lab	>60 dB	<0.1 nT	Below NV T ₂ limit
Acoustic	~40 dB SPL (quiet lab)	>80 dB combined	<–40 dB SPL	Negligible
Vibration (1–100 Hz)	~10 ^{–9} m/s ² lab floor	>60 dB (active+passive)	<10 ^{–11} m/s ²	Negligible
Thermal drift	±1 K typical lab	Active control	±0.01 K	Below NV T ₂ limit
Cosmic rays	~1/cm ² /min at surface	None (Phase 0)	~1/cm ² /min	Flagged in data

Cosmic Ray Note
Phase 0 (surface location) does not attenuate cosmic rays. Cosmic ray events are flagged in quantum sensor data by coincidence with scintillator veto counter (optional addition, ~€5K). Phase 1 underground location provides >10⁶ cosmic ray attenuation via rock overburden.

END OF DOCUMENT

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